

Use-Case Flow Specification

Use Case: Identify Recurring Subscriptions

Primary Actor: Individual User

Goal: To identify recurring subscriptions from the user's imported transaction data and display their subscription details.

Preconditions

1. The user has access to the Personal Subscription & Recurring Expense Auditor.
2. The user has successfully imported financial transaction data into the system.
3. The imported transaction data is in a supported CSV or OFX format.

Postconditions

1. Recurring subscription transactions have been identified by the system.
2. The identified subscriptions and their relevant details are available for the user to view.
3. The system has calculated the expected renewal date and recurring payment amount where applicable.

Main Success Scenario

1. The Individual User selects "**Identify Recurring Subscriptions**".
2. The system validates the imported transaction data.
3. The system reads and processes the transaction records.
4. The system groups transactions based on merchant and payment information.
5. The system analyzes the frequency and amount of transactions to identify recurring billing patterns.
6. The system identifies transactions that match recurring subscription patterns.
7. The system determines the expected renewal date and recurring payment amount for each identified subscription.
8. The system stores the identified subscription information.
9. The system displays the identified recurring subscriptions to the user.
10. The use case ends successfully.

Alternate Flow

A1. Invalid or Unreadable Transaction Data

1. At Step 2, the system detects that the imported transaction data is invalid or cannot be read.
2. The system displays an error message indicating that the transaction data could not be processed.

3. The user is prompted to import a valid CSV or OFX transaction file.
4. The user may import the transaction data again.
5. If valid data is provided, the use case continues from Step 2 of the Main Success Scenario.
6. If the user does not provide valid data, the use case ends without identifying subscriptions.

A2. No Recurring Subscriptions Found

1. At Step 5, the system does not find any transaction pattern that meets the criteria for a recurring subscription.
2. The system informs the user that no recurring subscriptions were identified.
3. The system displays the available transaction information.
4. The use case ends.